# Brainstem and Cranial Nerves - II

Aftab Merchant, MBBS, FCPS, MHPE
Professor of Medical Education
Assistant Dean of Anatomical Education
Kirk Kerkorian School of Medicine
University of Nevada Las Vegas

# Learning Objectives (2026)

1. Recognize the characteristic neurological signs and symptoms associated with lesions involving brainstem nuclei, cranial nerve fascicles, long tracts, and vascular territories. (CLO 3)
2. Apply knowledge of brainstem anatomy and associated neurological deficits to localize common brainstem syndromes. (CLO 3)

**Pre-Class Assignment:**

- Preview the PowerPoint slides. You are not expected to memorize details before class.
- Review the major brainstem and cranial nerve concepts from Session 3 – Brainstem and Cranial Nerves I, with particular attention to cranial nerve nuclei, long tracts, and vascular territories.
- Watch: Rule of 4s (15 min)

# Blood Supply – Circle of Willis

## Anterior Circulation

- Internal carotid artery → ACA, MCA

## Posterior Circulation

- Vertebral arteries → PICA
- Vertebral arteries unite → Basilar artery
- Basilar artery branches → AICA, SCA, pontine aa.
- Basilar artery terminates as → PCA

## Communicating Arteries

- Anterior communicating artery (ACOM)
- Posterior communicating arteries (PCOM)

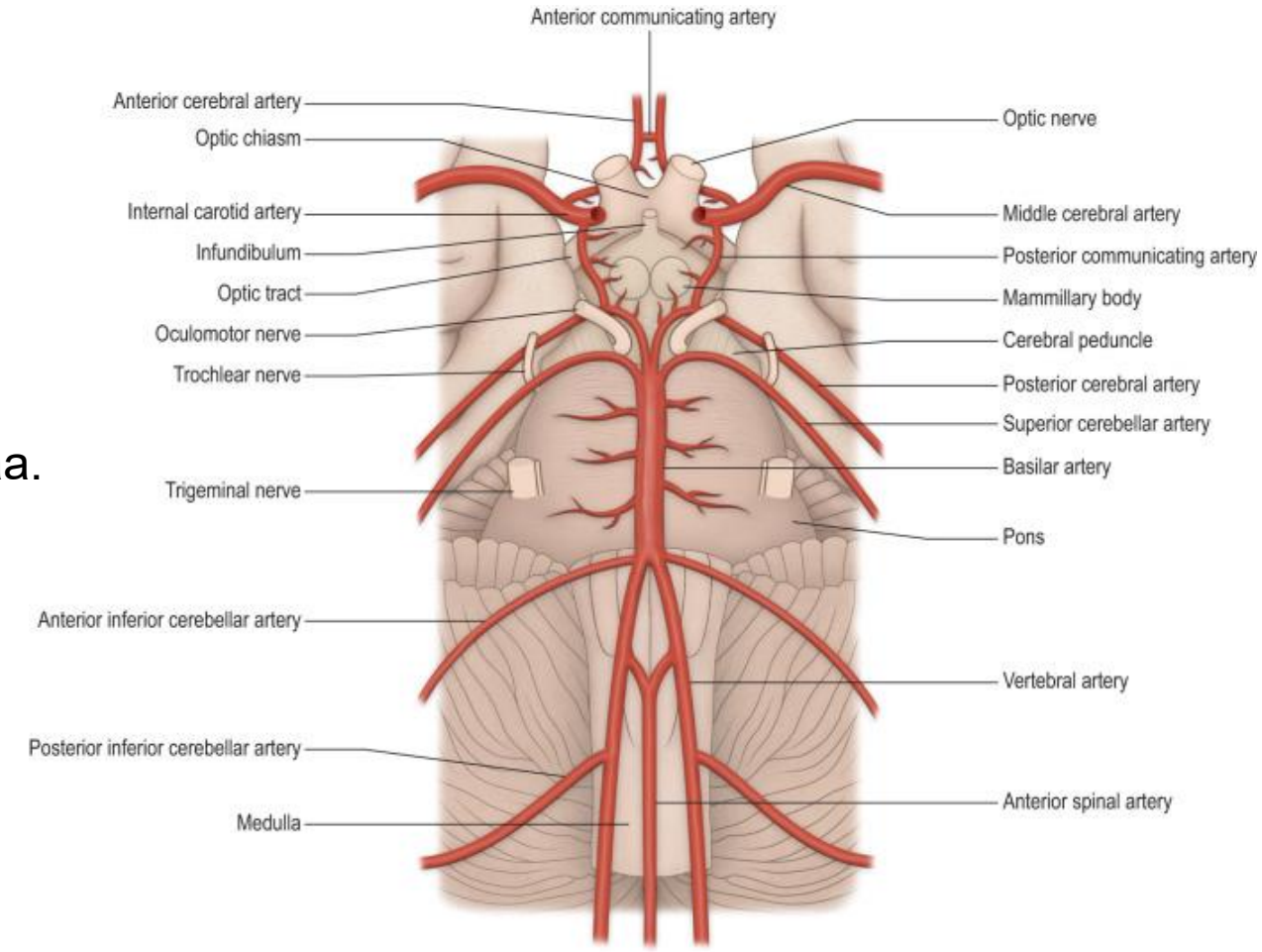



Fig. 7.2 The arrangement of arterial vessels on the base of the brain. The diagram shows the circulus arteriosus (circle of Willis).
Crossman, Neuroanatomy: An Illustrated Color Text, 6e.

# Major Arterial Supply

## Midbrain

- Posterior cerebral artery (PCA)
- Superior cerebellar artery (SCA)

## Pons

- Pontine branches of basilar artery
- Anterior inferior cerebellar artery (AICA)
- Superior cerebellar artery (SCA)

## Medulla

- Vertebral artery
- Posterior inferior cerebellar artery (PICA)
- Anterior spinal artery

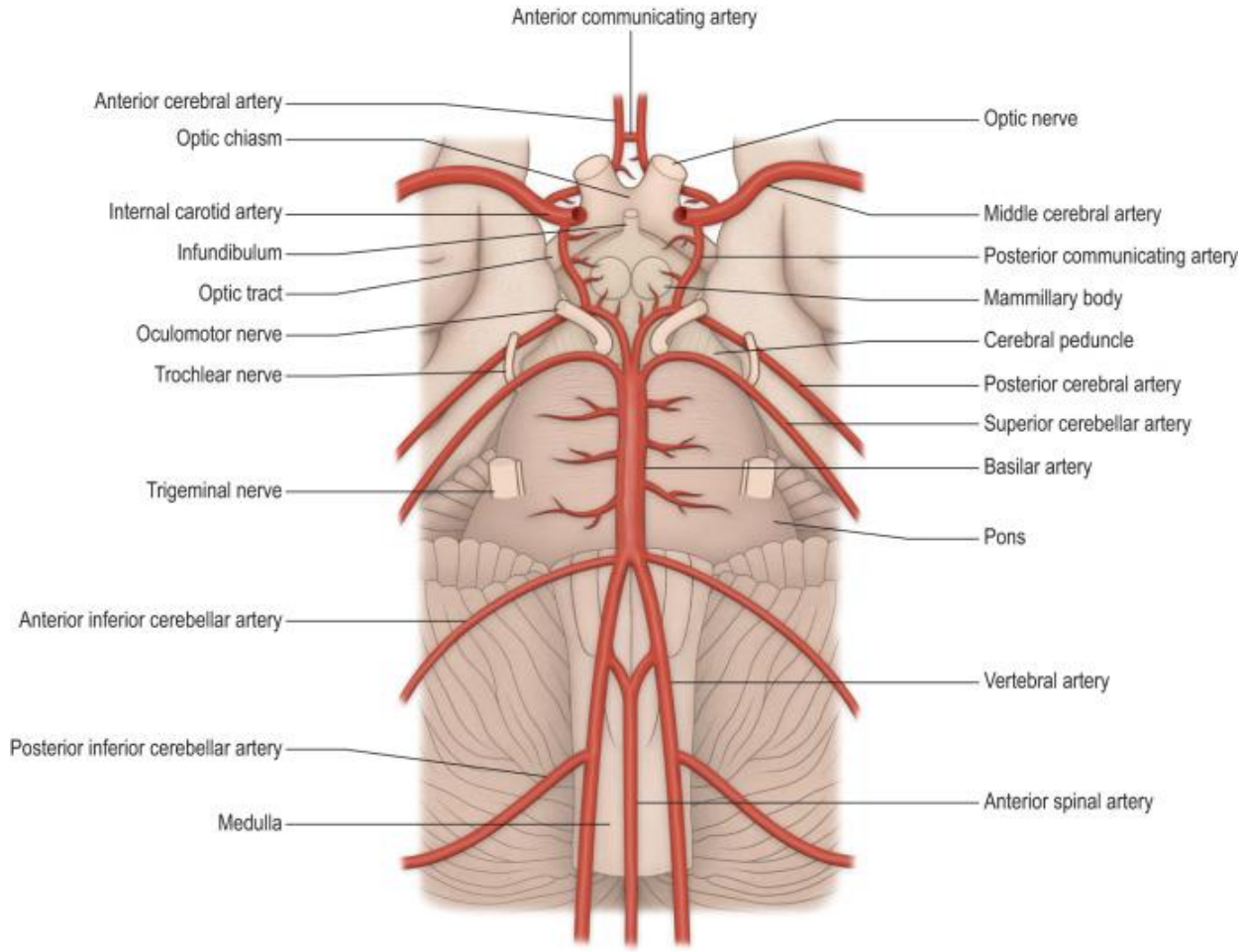
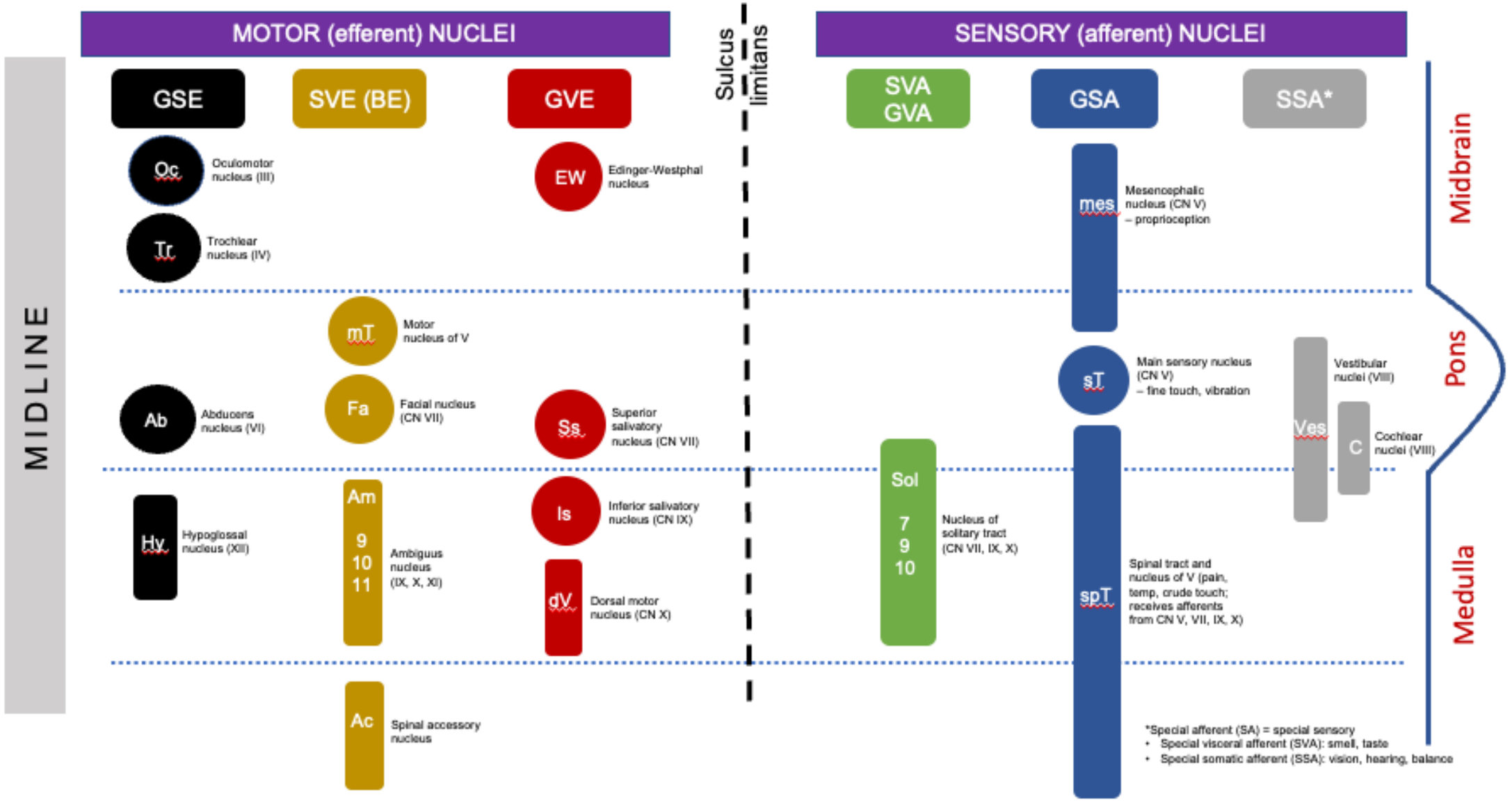



Fig. 7.2 The arrangement of arterial vessels on the base of the brain. The diagram shows the circulus arteriosus (circle of Willis).
Crossman, Neuroanatomy: An Illustrated Color Text, 6e.

# Cranial Nerve Nuclei in the Brainstem



# Brainstem Localization: A Four-Step Approach

1. **Cranial nerve deficit → LEVEL**
   - Midbrain → III, IV
   - Pons → V–VIII
   - Medulla → IX, X, XII

2. **Body deficit → LONG TRACT**
   - Weakness → corticospinal
   - Vibration/proprioception → medial lemniscus
   - Pain/temperature → spinothalamic

3. **Determine LOCATION**
   - Medial vs lateral

4. **Identify VASCULAR TERRITORY**
   - Match the involved structures to the artery



CN → Level | Body → Tract | Medial/Lateral → Location | Artery → Syndrome

# The Crossed Brainstem Pattern

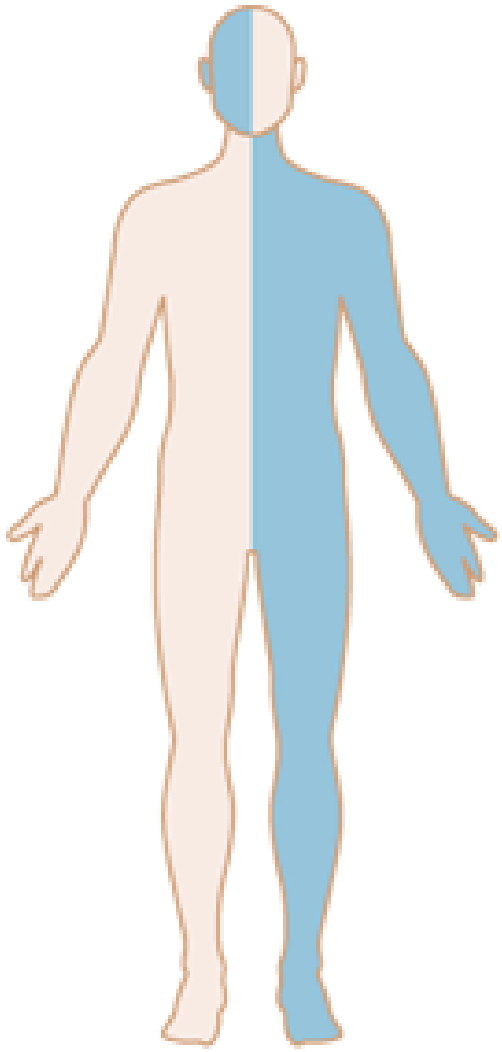
## Brainstem Lesions: Face vs Body

- Cranial nerve nuclei/fascicles affect the ipsilateral head/face
- Long tracts may produce contralateral body deficits
- Therefore, brainstem lesions commonly produce crossed neurological findings

## Example

- Ipsilateral facial pain/temperature loss
- Contralateral body pain/temperature loss

→ Brainstem lesion



Brainstem Lesion

Ipsilateral cranial nerve/face deficit + contralateral body deficit → think BRAINSTEM



Berkowitz. *Clinical Neurology and Neuroanatomy*. McGraw Hill.

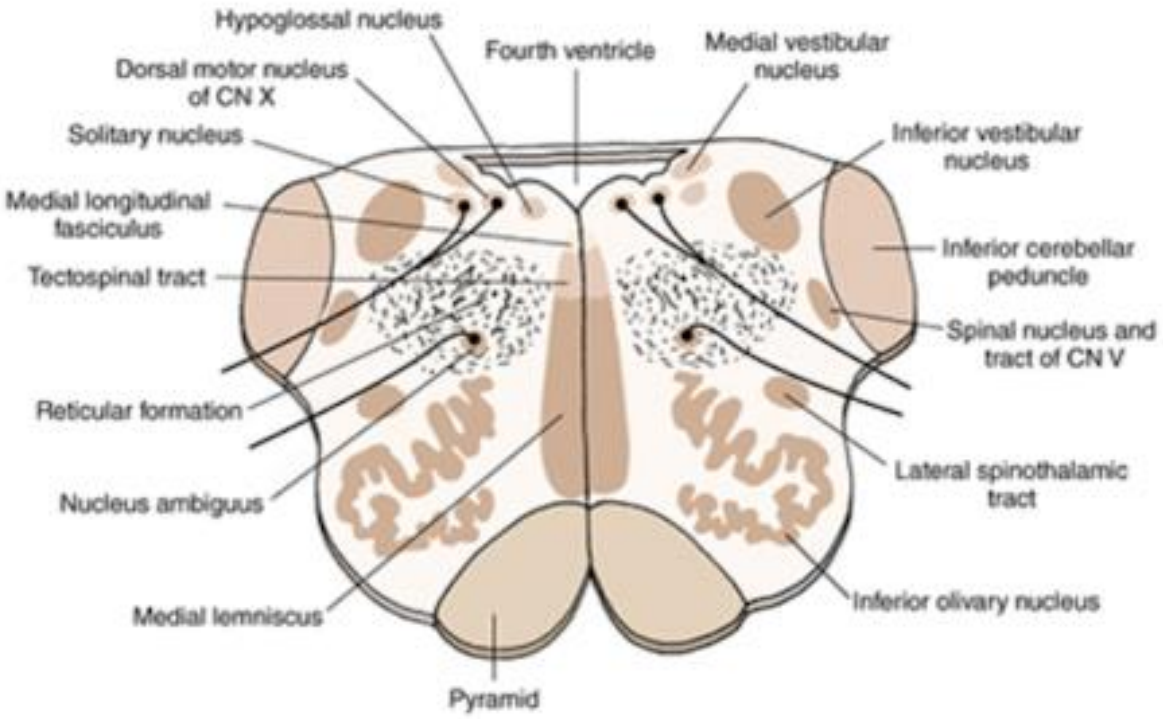
# Medial vs Lateral Localization

## Medial

- Corticospinal tract → motor
- Medial lemniscus → vibration/proprioception
- MLF → conjugate eye movements
- Medially located motor CN nuclei/fascicles

## Lateral

- Spinothalamic tract → body pain/temperature
- Spinal trigeminal system → facial pain/temperature
- Cerebellar pathways → coordination
- Descending sympathetic fibers → sympathetic function
- Selected sensory/autonomic CN nuclei




Central Medulla

Medial → motor + medial lemniscus | Lateral → pain/temp + cerebellar/autonomic



Fig. 9.8. Siegel, EN, 4e.

# Horner Syndrome

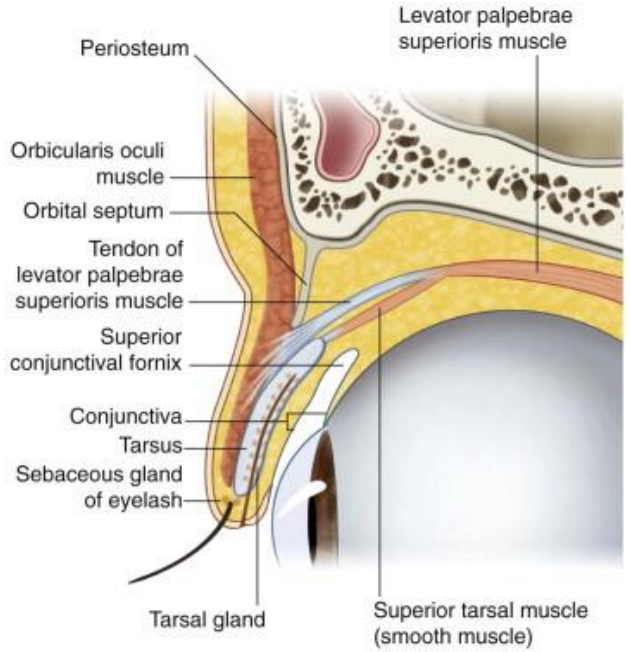
**Cause:** Interruption of sympathetic innervation to the head

## Classic Findings

- Miosis → loss of pupillary dilator
- Mild ptosis → loss of superior tarsal muscle
- Anhidrosis → loss of sympathetic innervation to sweat glands

## Brainstem Localization

- Descending hypothalamospinal sympathetic fibers travel through the lateral brainstem
- Lateral brainstem lesion → ipsilateral Horner syndrome






Ptosis: CN III Palsy



Partial Ptosis: loss of superior tarsal innervation

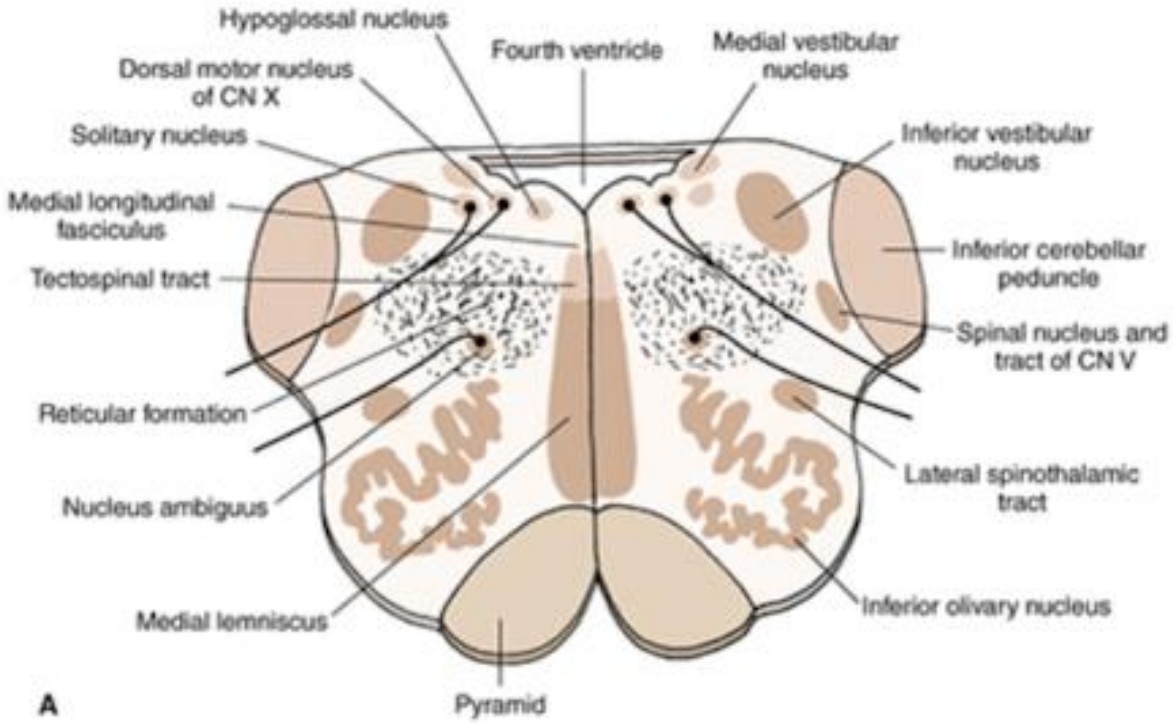


Fig. 8.74 Gray's Anatomy for Students, 3e.
Image: American Academy of Ophthalmology website
Image: Wikipedia

# Medial Medulla: Localize the Structures

## Identify three structures:

- Pyramid / corticospinal tract → body motor
- Medial lemniscus → vibration/proprioception
- CN XII fascicle → tongue motor




Central Medulla

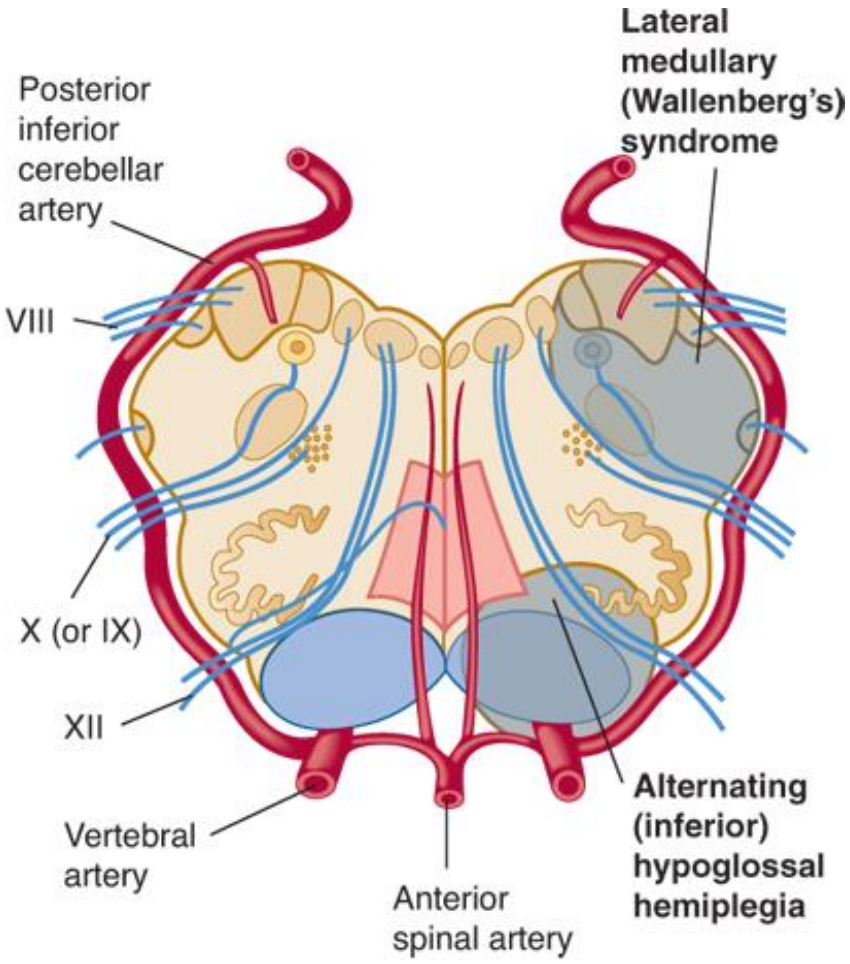
If these three structures were injured together, what would you predict?



Fig. 9.8. Siegel, EN, 4e.

# Medial Medullary Syndrome (Dejerine Syndrome)

| Structure | Deficit |
|---|---|
| Corticospinal tract | Contralateral body weakness |
| Medial lemniscus | Contralateral loss of vibration/proprioception |
| CN XII fascicle | Ipsilateral tongue weakness → deviation toward lesion |
| Vascular territory: anterior spinal or vertebral artery branches | |




**Ipsilateral XII deficit + contralateral motor/proprioception loss → medial medulla**

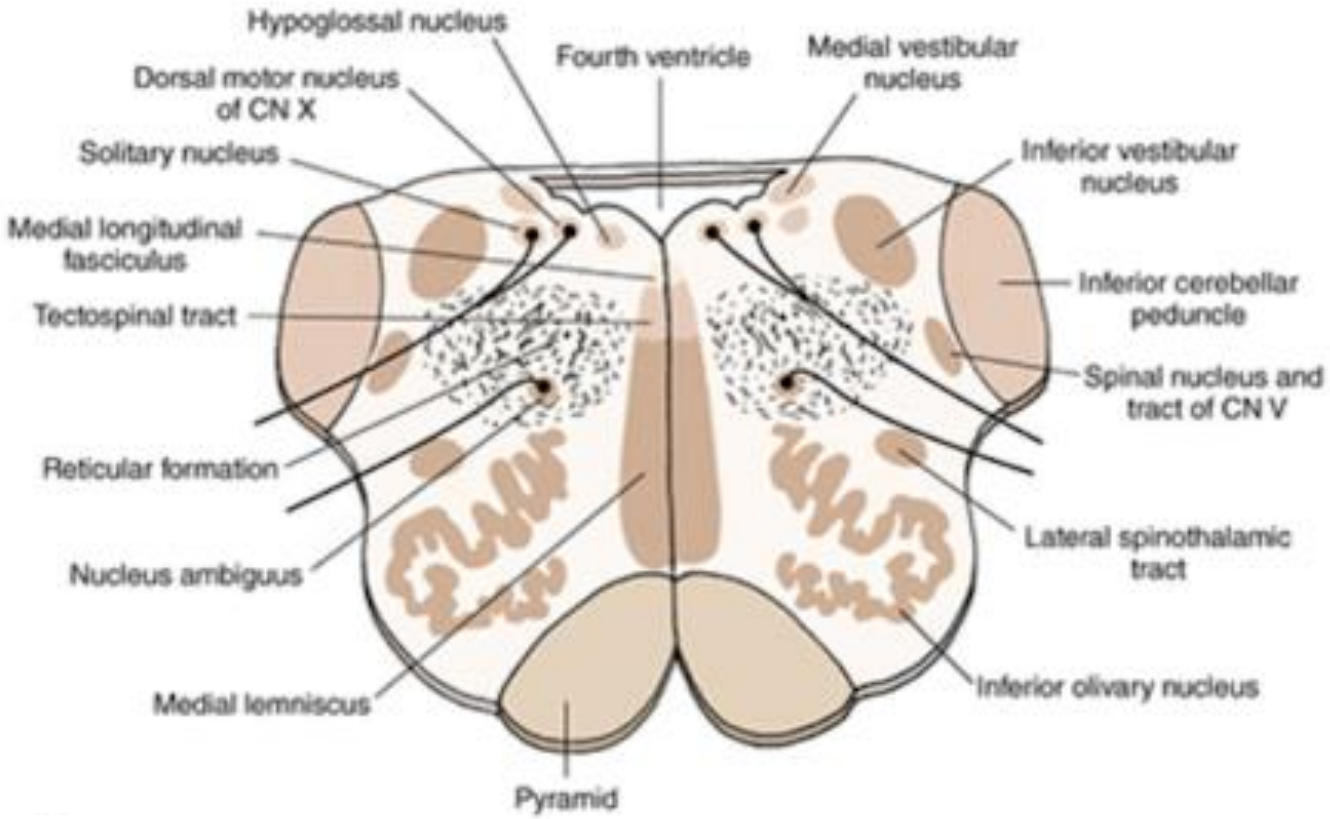


Waxman, 28e.

# Lateral Medulla: Localize the Structures

**Focus only on:**

- Nucleus ambiguus
- Spinal trigeminal nucleus/tract
- Spinothalamic tract
- Inferior cerebellar peduncle
- Vestibular nuclei
- Descending sympathetic fibers




Central Medulla

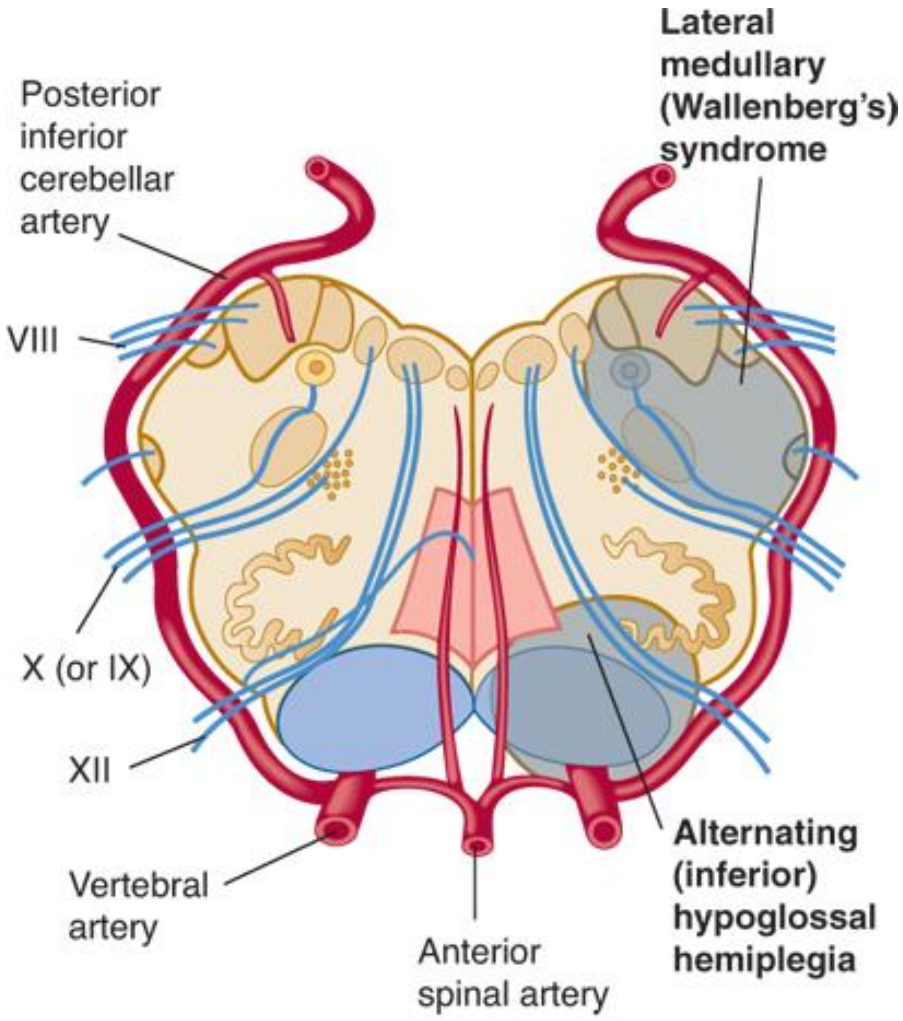
Swallowing/voice + crossed pain/temp + ataxia/autonomic findings → lateral medulla



Fig. 9.8. Siegel, EN, 4e.

# Lateral Medullary (Wallenberg) Syndrome

| Structure | Deficit |
|---|---|
| Nucleus ambiguus | Dysphagia, hoarseness |
| Spinal trigeminal system | Ipsilateral facial pain/temp loss |
| Spinothalamic tract | Contralateral body pain/temp loss |
| Inferior cerebellar peduncle | Ipsilateral ataxia |
| Vestibular nuclei | Vertigo, nystagmus |
| Descending sympathetics | Ipsilateral Horner syndrome |
| Vascular territory: PICA or vertebral artery | |




Nucleus ambiguus
involvement is the key clue to lateral medulla

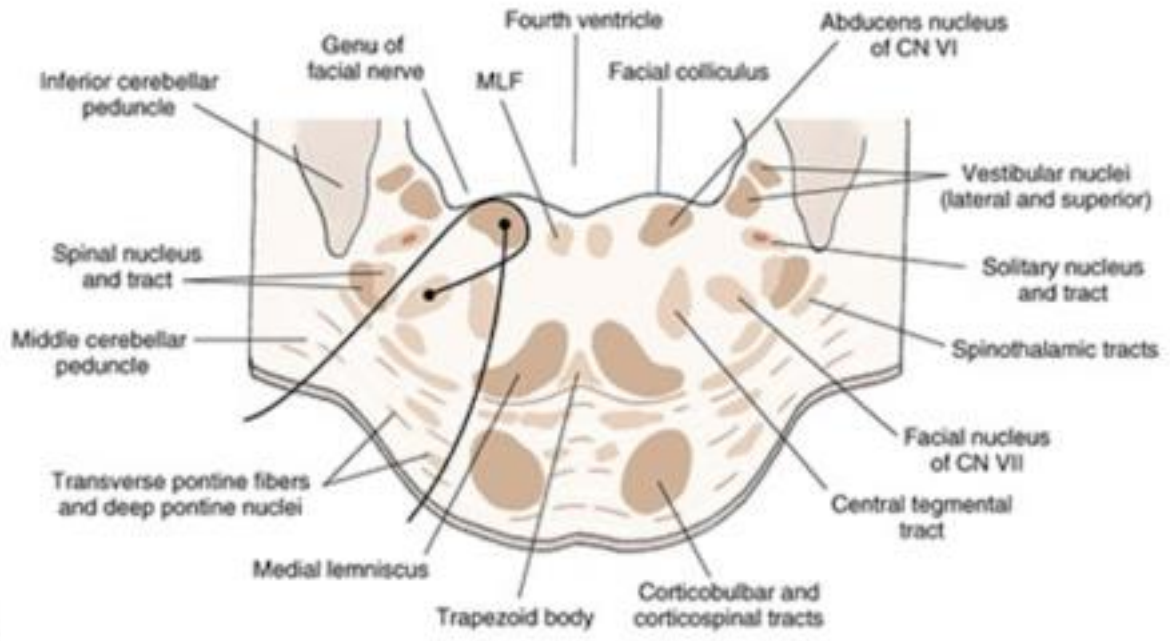


Waxman, 28e.

# Medial Pons: Localize the Structures

## Identify three structures:

- Corticospinal tract → body motor
- Medial lemniscus → vibration/proprioceptior
- CN VI fascicle → lateral rectus motor




Caudal Pons

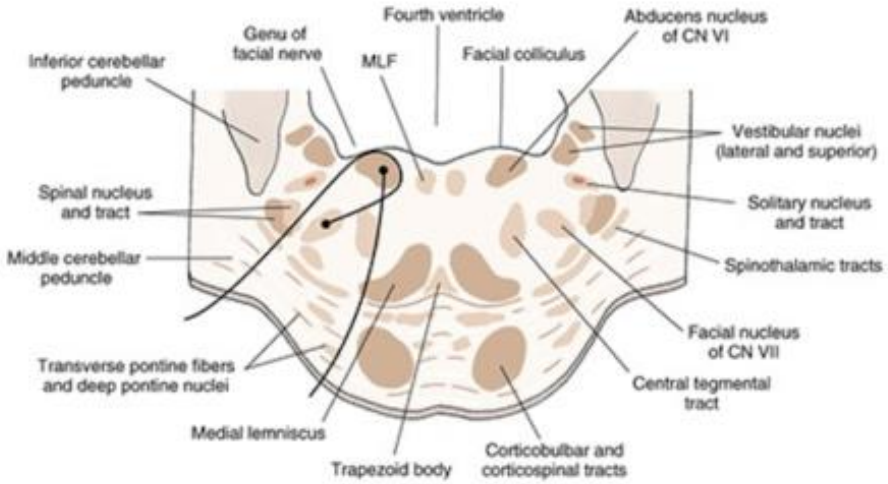
If these three structures were injured together, what would you predict?



Fig. 10.5. Siegel, EN, 4e.

# Medial Pontine Syndrome

| Structure | Deficit |
|---|---|
| Corticospinal tract | Contralateral body weakness |
| Medial lemniscus | Contralateral loss of vibration/proprioception |
| CN VI fascicle | Ipsilateral impaired eye abduction |
| Vascular territory: paramedian branches of the basilar artery | |




Caudal Pons

Ipsilateral CN VI + contralateral motor/proprioception → medial pons

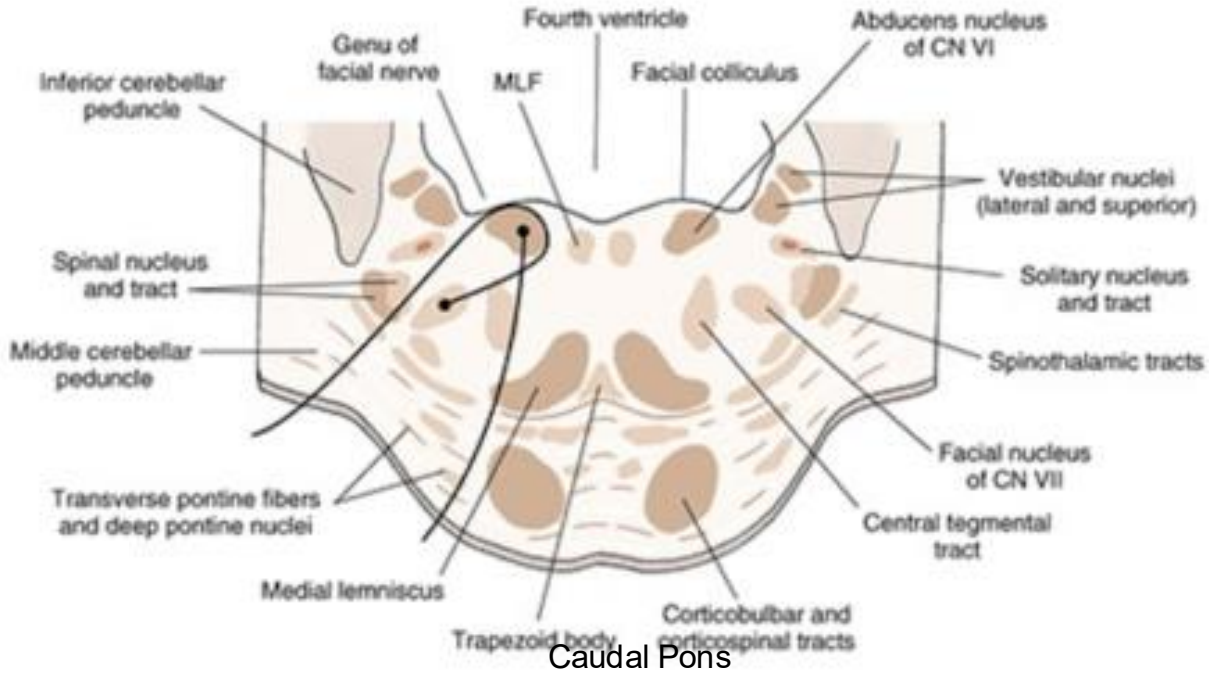


Fig. 10.5. Siegel, EN, 4e.

# Lateral Pons – Localize the Structures

## Key structures

- Facial nucleus/fascicle (CN VII)
- Vestibular/cochlear structures (CN VIII)
- Spinal trigeminal system
- Spinothalamic tract
- Cerebellar connections
- Descending sympathetic fibers



Caudal Pons

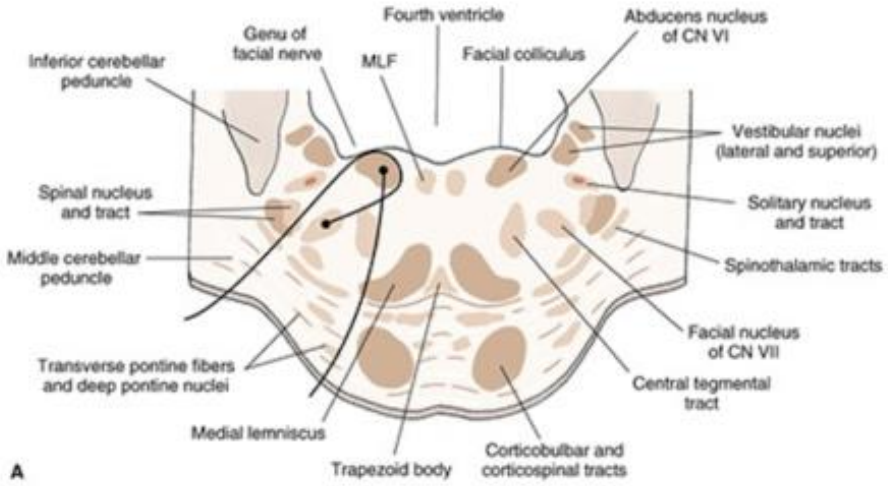
CN VII/VIII findings → think PONS



Fig. 10.4. Siegel, EN, 4e.

# Lateral Pontine Syndrome

| Structure | Deficit |
|---|---|
| Facial nucleus/fascicle (CN VII) | Ipsilateral facial paralysis |
| Vestibular nuclei (CN VIII) | Vertigo, nystagmus |
| Cochlear nuclei/pathway (CN VIII) | Ipsilateral hearing loss |
| Spinal trigeminal nucleus/tract | Ipsilateral facial pain/temp loss |
| Spinothalamic tract | Contralateral body pain/temp loss |
| Middle/inferior cerebellar peduncle | Ipsilateral ataxia |
| Descending sympathetic fibers | Ipsilateral Horner syndrome |
| Vascular territory: AICA | |




Caudal Pons

Facial paralysis → AICA/pons | Dysphagia + hoarseness → PICA/medulla



Fig. 10.5. Siegel, EN, 4e.

# Lateral Pons vs Lateral Medulla

| | AICA — Pons | PICA — Medulla |
|---|---|---|
| Level clue | **CN VII/VIII** | **CN IX/X** |
| Facial paralysis | **Yes** | No |
| Dysphagia/hoarseness | No/less characteristic | **Yes** |
| Ipsilateral face pain/temp ↓ | Yes | Yes |
| Contralateral body pain/temp ↓ | Yes | Yes |
| Ataxia/vertigo | Yes | Yes |

Facial weakness → AICA | Hoarseness/dysphagia → PICA

# Medial Midbrain Localization

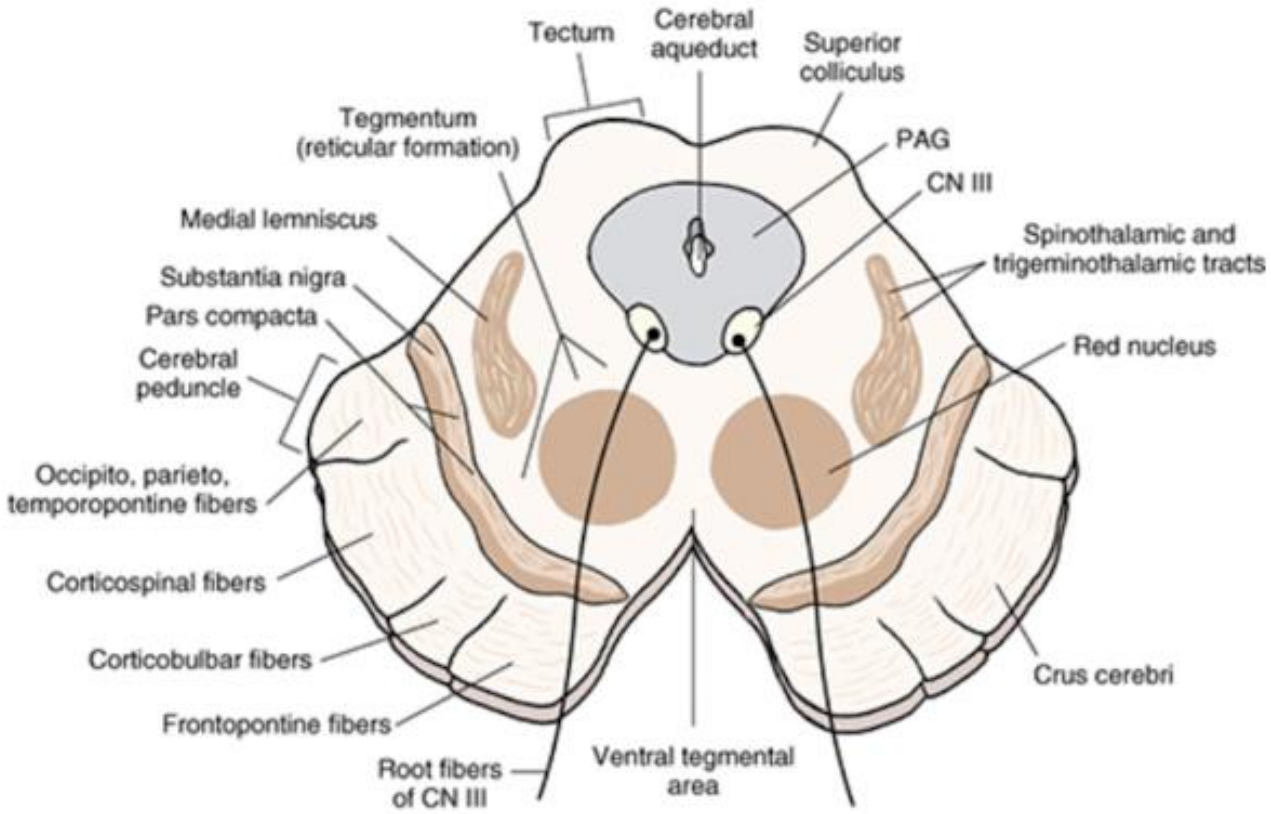
## Key structures

- CN III fascicles
- Cerebral peduncle / corticospinal tract

## Predict the findings:

- CN III → ipsilateral oculomotor palsy
- Corticospinal → contralateral body weakness

**CN III + contralateral weakness → medial midbrain**



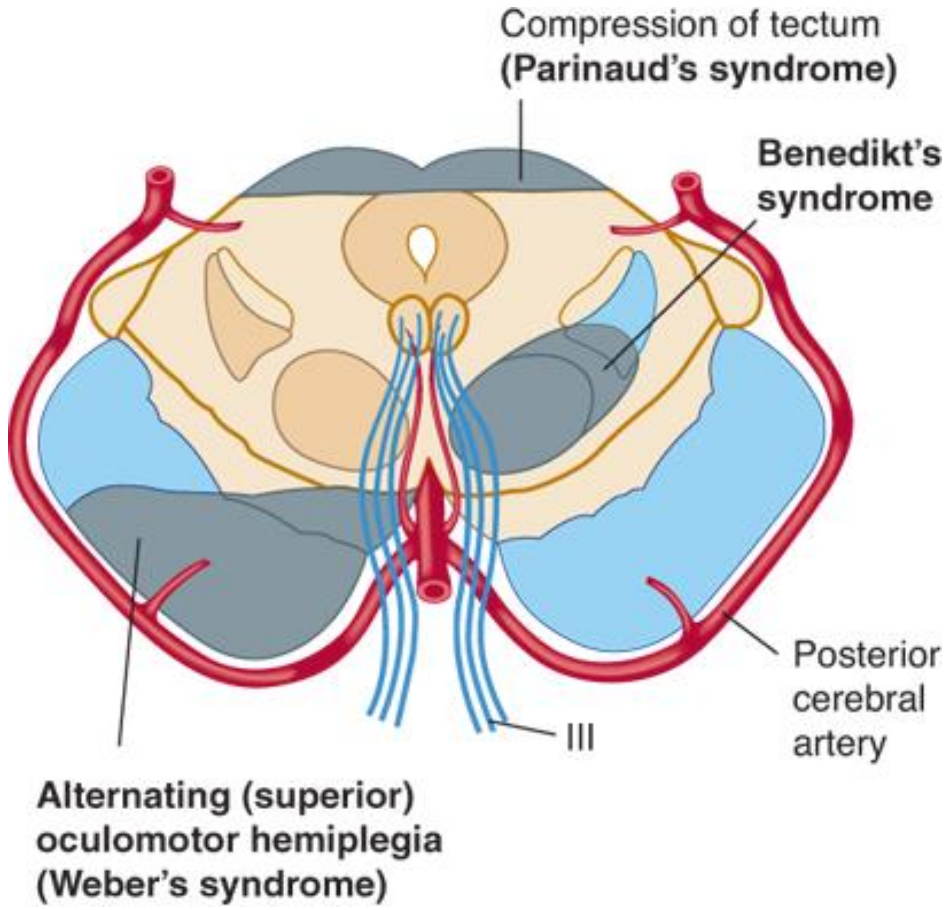

Midbrain – superior colliculus level



Fig. 11.5. Siegel, EN, 4e.

# Weber Syndrome

| Structure | Deficit |
|---|---|
| CN III fascicle | Ipsilateral CN III palsy |
| Corticospinal tract | Contralateral body weakness |
| Corticobulbar tract | Contralateral lower facial weakness |
| **Location:** ventromedial midbrain | |
| **Vascular territory:** PCA branches | |



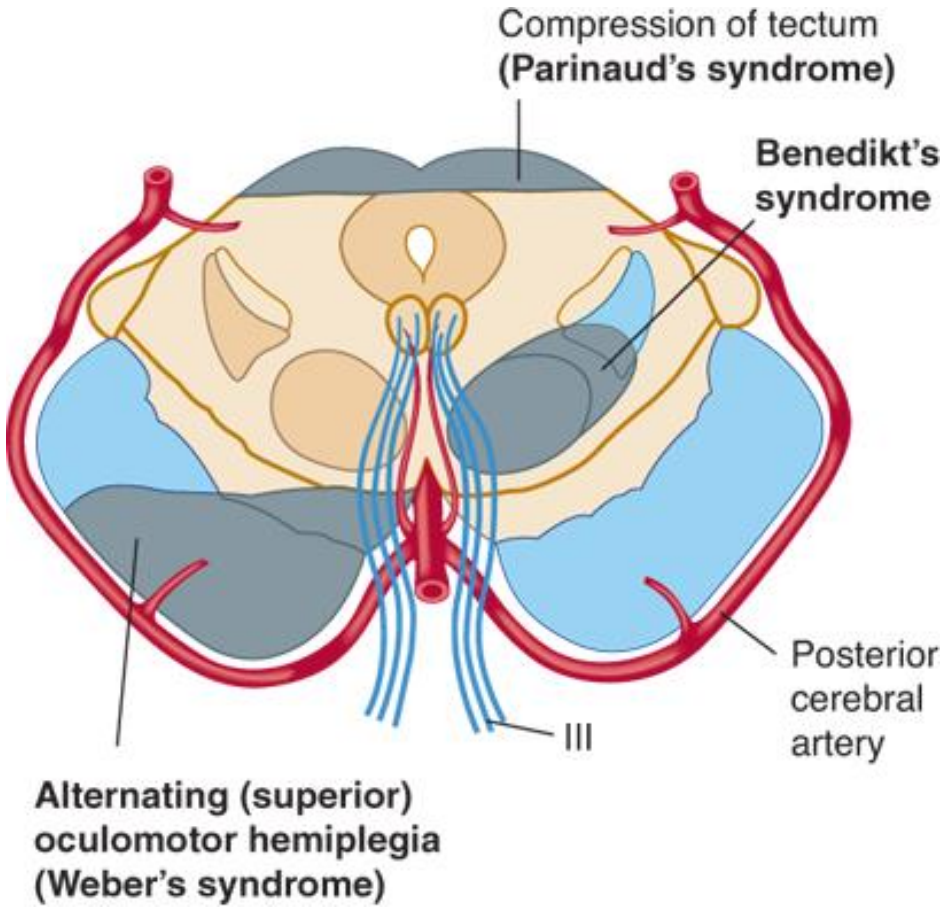

Midbrain – superior colliculus level, Waxman, 28e.

Ipsilateral CN III + contralateral motor deficits → medial midbrain

# Dorsal Midbrain (Parinaud) Syndrome

- Location: dorsal midbrain / pretectal region
- Impaired upward gaze *(vertical gaze palsy)*
- Pupils constrict to near, but not to light *(light-near dissociation)*
- Convergence-retraction nystagmus
- Often associated with compression by a pineal region mass

Vertical gaze abnormality → dorsal midbrain




Midbrain – superior colliculus level, Waxman, 28e.

# Brainstem Localization

1. **Find the cranial nerve deficit → LEVEL**
2. **Find the body deficit → LONG TRACT**
3. **Combine the structures → MEDIAL vs LATERAL**
4. **Identify the vascular territory → ARTERY**
5. **Only then → NAME THE SYNDROME**

Don't memorize syndromes as lists; localize the anatomy

# Brainstem and Cranial Nerves – II

Aftab Merchant, MBBS, FCPS, MHPE
Professor of Medical Education
Assistant Dean of Anatomical Education
Course Director Anatomy and Neuroscience
Tel: (702) 545-2050 | aftab.merchant@unlv.edu
Schedule an Appointment

KIRK KERKORIAN SCHOOL OF MEDICINE | UNLV